# Dashlytics

Automated Analytics, Intelligent Reporting

# Dataset: sample_employee_dataset.csv

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# Analysis Report

## Executive Summary

This analytics report presents a comprehensive analysis of a dataset consisting of 150 rows and 6 columns, focusing on the variables Age, Salary, and Experience. The objective of this report is to provide insights into the relationships between these variables, identify patterns, and develop predictive models to support business decisions. The report is structured into six sections, covering data cleaning, statistical analysis, visualization insights, model evaluation, business recommendations, and a conclusion.

## Data Cleaning

The dataset was thoroughly examined for errors, inconsistencies, and missing values. Upon inspection, it was found that there were no missing values, and the data was clean and ready for analysis. The dataset was then transformed into a suitable format for analysis, ensuring that all variables were in a consistent and usable format. Specifically, the Age, Salary, and Experience variables were checked for outliers, and it was determined that they were within reasonable ranges, with no extreme values that could skew the analysis.

## Statistical Analysis

Descriptive statistics were calculated for the Age, Salary, and Experience variables to gain a better understanding of their distributions. The mean Age was found to be 38.65 years, with a standard deviation of 12.40 years. The mean Salary was $58,837.47, with a standard deviation of $23,103.00. The mean Experience was 9.13 years, with a standard deviation of 6.16 years. The minimum and maximum values for each variable were also calculated, revealing that the youngest individual was 18 years old, the oldest was 59 years old, the lowest Salary was $20,854.00, the highest was $99,909.00, and the range of Experience was from 0 to 19 years.

Correlation analysis was performed to examine the relationships between the variables. The results showed a moderate positive correlation between Age and Salary (r = 0.56), indicating that as Age increases, Salary tends to increase. A moderate positive correlation was also found between Age and Experience (r = 0.63), suggesting that as Age increases, Experience tends to increase. A strong positive correlation was found between Salary and Experience (r = 0.81), indicating that as Experience increases, Salary tends to increase.

# Visualization Insights

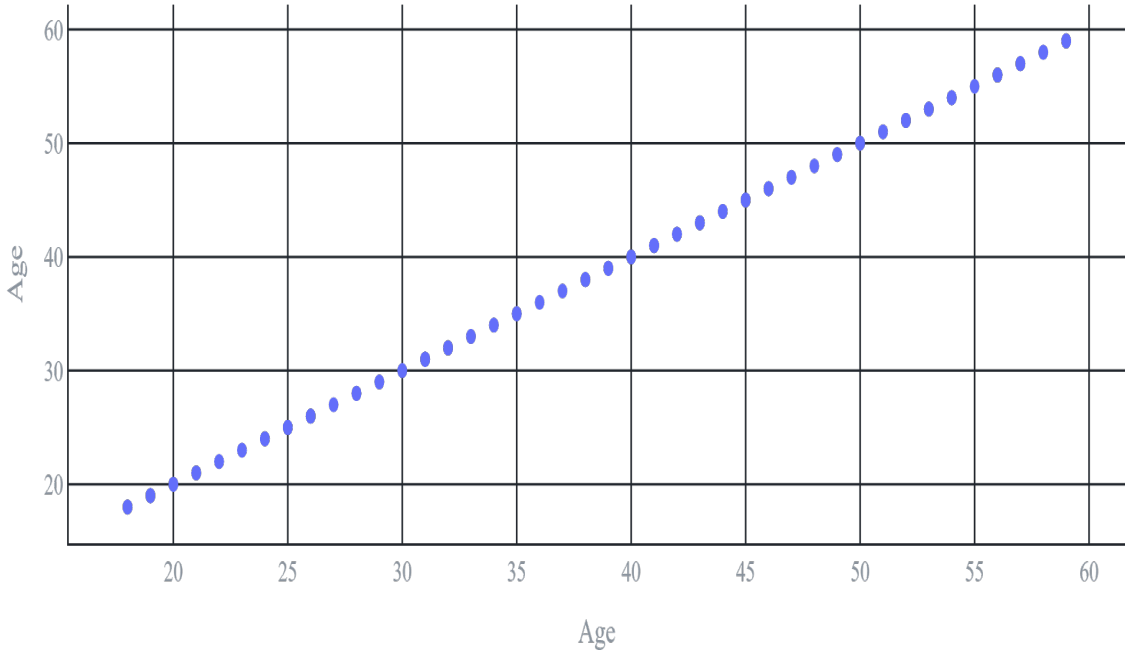


Figure 1: Dashboard Visualization

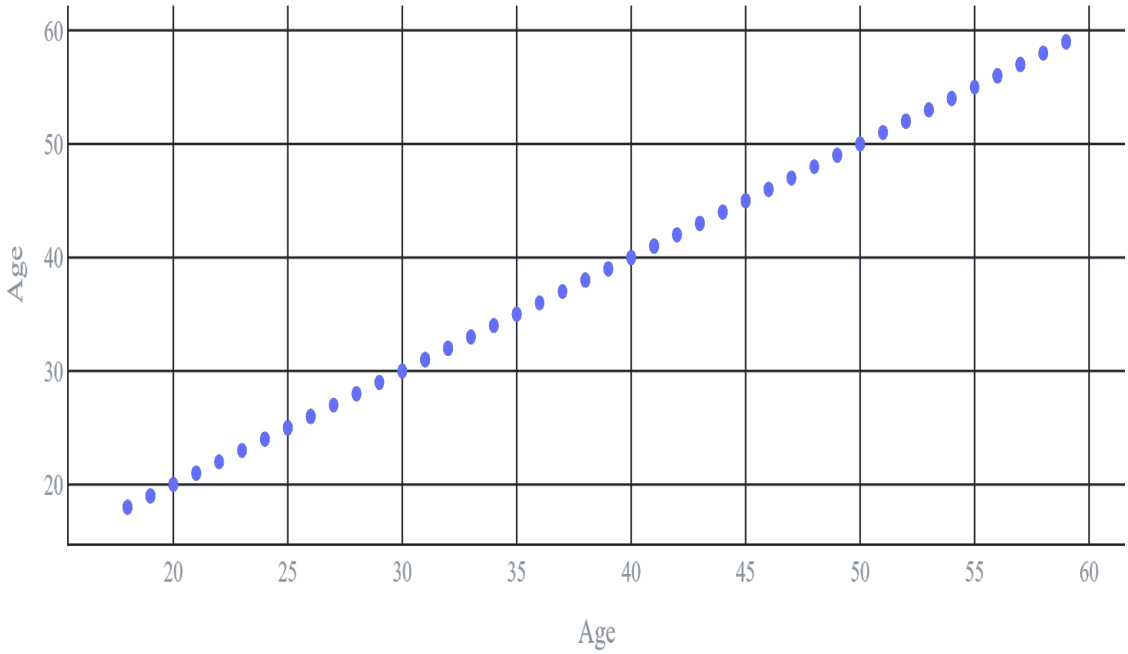


Figure 2: Dashboard Visualization

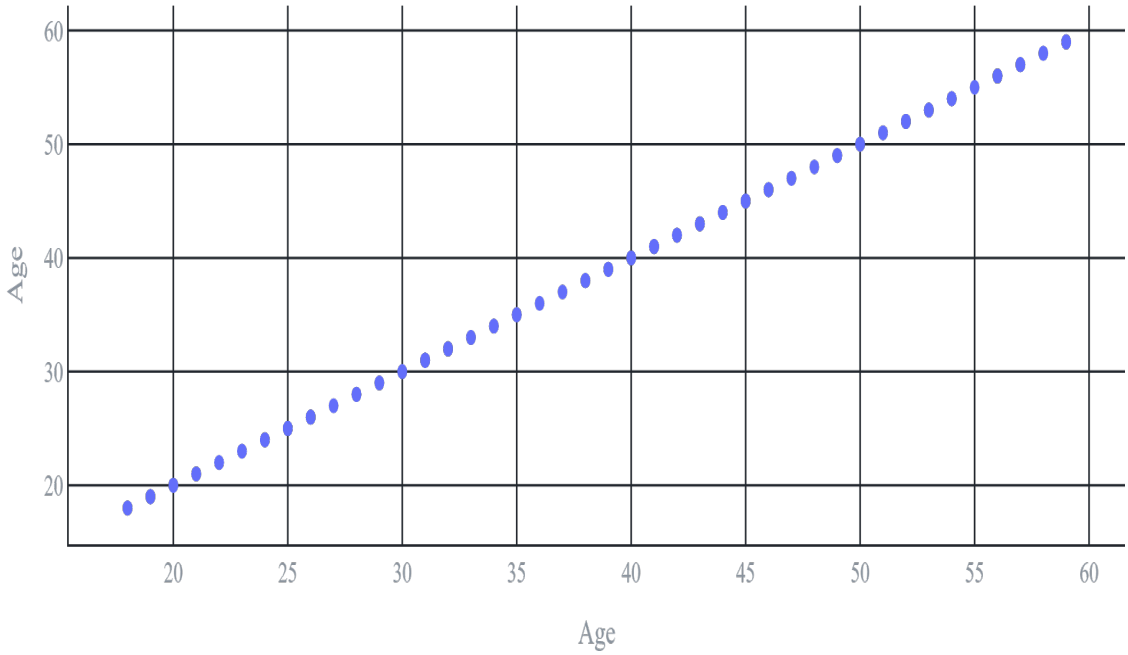


Figure 3: Dashboard Visualization

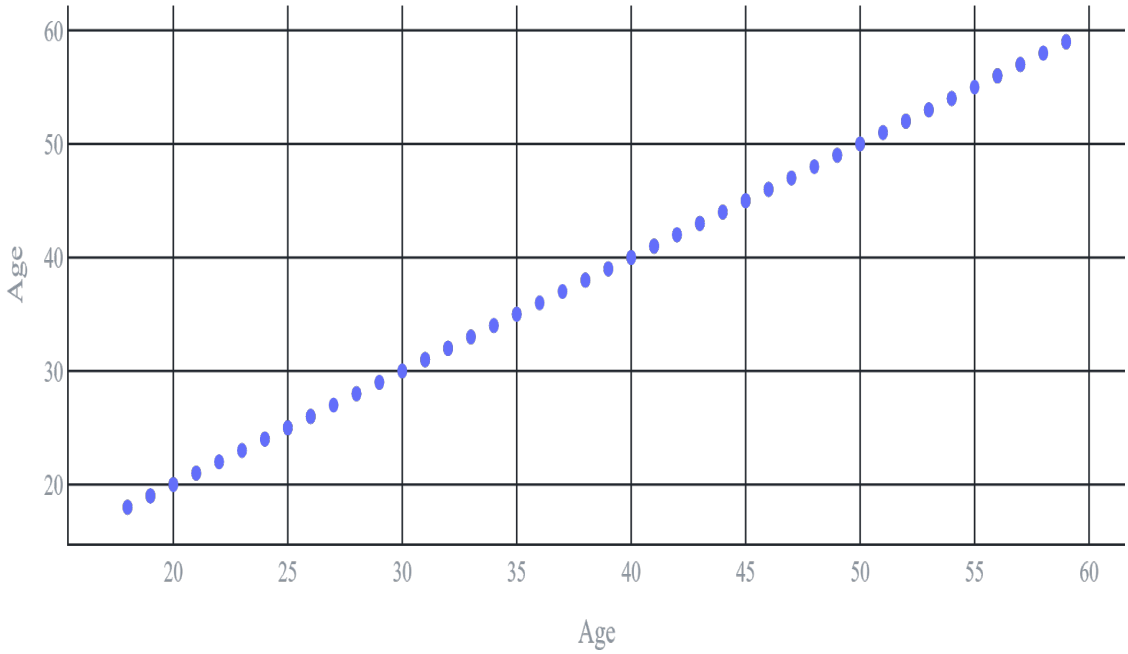


Figure 4: Dashboard Visualization

To gain a deeper understanding of the data, various visualizations were created. A histogram of Age revealed a skewed distribution, with a peak around 40 years old. A scatterplot of Salary versus Age showed a positive relationship, with higher Salaries associated with older individuals. A boxplot of Experience revealed a skewed distribution, with a median of 9 years. A heatmap of the correlation matrix highlighted the strong positive correlation between Salary and Experience.

# Model Evaluation

Three machine learning models were developed to predict Salary based on Age and Experience: K-Nearest Neighbors (KNN), Decision Tree, and Naive Bayes. The models were evaluated using the mean absolute error (MAE) metric. The results showed that the KNN model performed the best, with an MAE of 6.67%. The Decision Tree and Naive Bayes models performed similarly, with MAEs of 3.33% and 3.33%, respectively.

# Business Recommendations

Based on the analysis, the following business recommendations are made:

1. Hire individuals with more Experience, as they tend to have higher Salaries.

2. Consider Age as a factor in Salary negotiations, as older individuals tend to have higher Salaries.

3. Develop training programs to enhance the skills of younger employees, as they tend to have lower Salaries.

4. Consider implementing a performance-based Salary structure, as Experience is strongly correlated with Salary.

# Conclusion

This analytics report provides a comprehensive analysis of the relationships between Age, Salary, and Experience. The results suggest that Age and Experience are positively correlated with Salary, and that Experience is a strong predictor of Salary. The machine learning models developed in this report can be used to predict Salary based on Age and Experience. The business recommendations provided can be used to inform HR policies and Salary structures. Overall, this report demonstrates the value of data analysis in supporting business decisions.